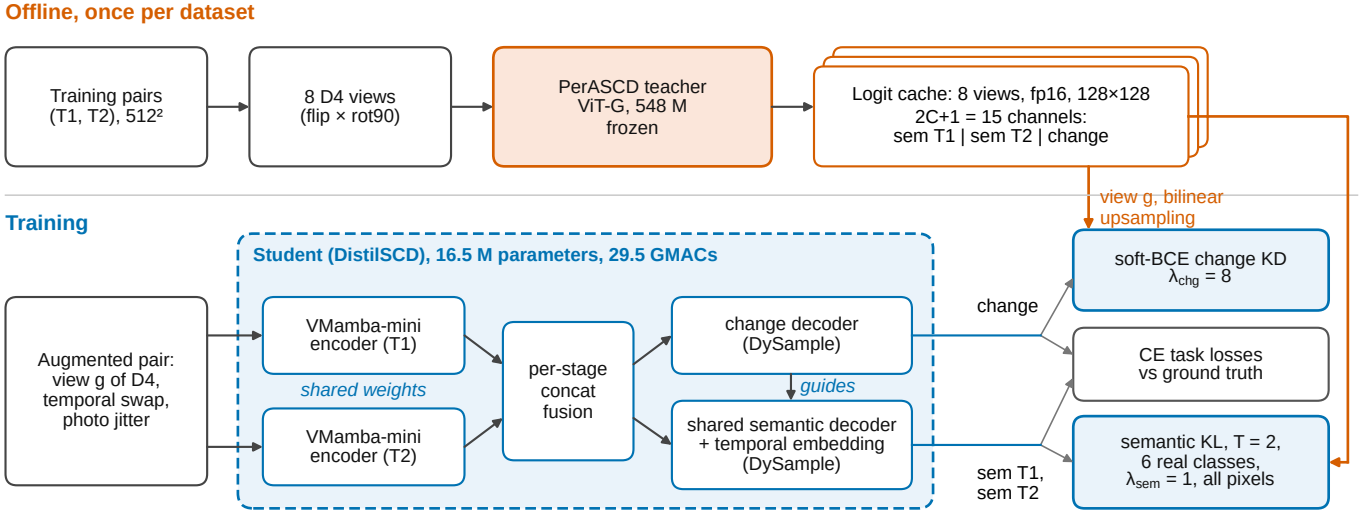
Offline, once per dataset
Training pairs (T1, T2), 512²
8 D4 views (flip × rot90)
PerASCD teacher ViT-G, 548 M frozen
Logit cache: 8 views, fp16, 128×128
2C+1 = 15 channels:
sem T1 | sem T2 | change
view g, bilinear upsampling
Training
Student (DistilSCD), 16.5 M parameters, 29.5 GMACs
Augmented pair: view g of D4, temporal swap, photo jitter
VMamba-mini encoder (T1)
shared weights
VMamba-mini encoder (T2)
per-stage concat fusion
change decoder (DySample)
guides
shared semantic decoder + temporal embedding (DySample)
change
sem T1, sem T2
soft-BCE change KD
λchg = 8
CE task losses vs ground truth
semantic KL, T = 2, 6 real classes,
λsem = 1, all pixels
Inference: the student alone; teacher, cache and losses are discarded.